

Experiment 2

AIM: Design Employee Payroll System using Inheritance, Interface & Polymorphism

Source: Practical 2.cs

Code:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace pract1.net
{
    interface IPayroll
    {
        void CalculateSalary();
        void DisplayDetails();
    }

    class Employee
    {
        public string Name;
        public int EmpId;
        public double BasicSalary;
        public void GetDetails()
        {
            Console.WriteLine("Enter Employee ID: ");
            EmpId =
            Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter Employee Name:
");
            Name = Console.ReadLine();
        }
    }
}
```

```
        Console.WriteLine("Enter Basic Salary: ");

        BasicSalary          =
        Convert.ToDouble(Console.ReadLine());
    }
}

class FullTimeEmployee : Employee, IPayroll
{
    double hra, da, netSalary;

    public void CalculateSalary()
    {
        hra = BasicSalary * 0.20;
        da = BasicSalary * 0.10;
        netSalary = BasicSalary + hra + da;
    }

    public void DisplayDetails()
    {
        Console.WriteLine("\n----- Full Time
Employee -----");

        Console.WriteLine("Employee ID : " +
EmpId);

        Console.WriteLine("Employee Name : "
+ Name);

        Console.WriteLine("Basic Salary : " +
BasicSalary);

        Console.WriteLine("HRA (20%) : " +
hra);

        Console.WriteLine("DA (10%) : " + da);

        Console.WriteLine("Net Salary : " +
netSalary);
    }
}

class PartTimeEmployee : Employee,
IPayroll
```

```
{
    double netSalary;
    public void CalculateSalary()
    {
        netSalary = BasicSalary;
    }
    public void DisplayDetails()
    {
        Console.WriteLine("\n----- Part Time
Employee -----");
        Console.WriteLine("Employee ID : " +
EmpId);
        Console.WriteLine("Employee Name : "
+ Name);
        Console.WriteLine("Salary : " +
netSalary);
    } }
class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Payroll System");
        Console.WriteLine("1. Full Time
Employee");
        Console.WriteLine("2. Part Time
Employee");
        Console.Write("Enter Choice: ");
        Int choice =
Convert.ToInt32(Console.ReadLine());
        if (choice == 1)
        {
            IPayroll emp = new
FullTimeEmployee();
            ((FullTimeEmployee)emp).GetDetails();
```

```
        emp.CalculateSalary();
        emp.DisplayDetails();
    }
    else if (choice == 2)
    {
        IPayroll emp = new
        PartTimeEmployee();
        ((PartTimeEmployee)emp).GetDetails();
        emp.CalculateSalary();
        emp.DisplayDetails();
    }
    else{
        Console.WriteLine("Invalid Choice!");
    }

    Console.ReadKey();
    }
}
```

Output:

```
Payroll System
1. Full Time Employee
2. Part Time Employee
Enter Choice: 1
Enter Employee ID: 22
Enter Employee Name: shrey
Enter Basic Salary: 100000

----- Full Time Employee -----
Employee ID : 22
Employee Name : shrey
Basic Salary : 100000
HRA (20%) : 20000
DA (10%) : 10000
Net Salary : 130000
```

```
Payroll System
1. Full Time Employee
2. Part Time Employee
Enter Choice: 2
Enter Employee ID: 25
Enter Employee Name: shrey
Enter Basic Salary: 50000

----- Part Time Employee -----
Employee ID : 25
Employee Name : shrey
Salary : 50000
```

GitHub URL: <https://github.com/shrey2250/.net-practical-2.git>